

Native DS-TWR speed-limit study

Throughput, reliability, and unfiltered static accuracy

Test date: July 26, 2026, EEST
Platform: 5 ESP32-S3 + DW3000 modules, Raspberry Pi 5
Branch: `experiment/ds-twr-native-performance`
Firmware baseline: `a97cdce-dirty`
Accelerated firmware: `27a28b0-dirty` + instrumentation patch
Configuration: M1 tag; M2, M3, M4, M5 anchors
Geometry: surveyed 3×3 m square, tag at the center
Environment: static, line of sight

Key result. The validated 17 ms profile delivered **58.02 complete positions/s**, **99.85% frame completeness**, and **3.35 cm 2D RMSE**: a **3.83×** speedup over the 64 ms reference without a measurable static-accuracy penalty. The tested scheduling limit was 13 ms, which delivered **74.68 positions/s** and **3.47 cm RMSE**, but with much less timing margin and about 9.4 times more failed exchanges than the 17 ms profile.

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1 Executive summary

This study asks how far the current native three-message DS-TWR implementation can be accelerated without sacrificing the static accuracy observed in the earlier 64 ms field test. The radio exchange remains strictly POLL-RESP-FINAL; no additional ranging frames, temporal filters, moving averages, or position smoothing were introduced.

The sweep covered frame periods from 64 ms down to 13 ms. Six 60 s screening blocks identified the useful region. The two fastest candidates were then validated in the interleaved order 13 ms V1-17 ms V1-13 ms V2-17 ms V2, with two 120 s blocks per profile. Together with the immutable 3×120 s reference, the eligible data contain 1,200.6 s of measurements, 218,617 ranges, and 54,214 complete positions.

Table 1: Main results. The 17 and 13 ms rows use pooled data from their two 120 s validation blocks.

Profile	Position rate	Speedup	Complete	RMSE	P95	Radio success
64 ms reference	15.15 Hz	1.00×	97.34%	3.25 cm	4.80 cm	n/a
21 ms screening	47.10 Hz	3.11×	99.89%	3.56 cm	5.03 cm	99.982%
17 ms balanced	58.02 Hz	3.83×	99.85%	3.35 cm	4.65 cm	99.966%
13 ms maximum	74.68 Hz	4.93×	98.71%	3.47 cm	4.83 cm	99.678%

Decision. The 13 ms profile is the measured throughput limit of the current millisecond-resolution scheduler. It remains usable in this test, but its 3 ms slots have little reserve for host-task jitter. The 17 ms profile is the recommended operating point: it preserves essentially all of the measured accuracy, loses only 0.15% of frames, and leaves substantially more timing headroom. The modules were left on this 17 ms profile.

2 System and protocol

2.1 Native three-frame DS-TWR

For each anchor, the tag transmits POLL, the anchor returns RESP using delayed transmission, and the tag sends FINAL. The final frame contains the tag-side hardware timestamps; the anchor already owns the complementary receive/transmit timestamps and therefore calculates the range without a report or acknowledgement frame. This is the native asymmetric double-sided two-way-ranging structure described by Decawave and used by the Qorvo examples [2, 1].

The time of flight is estimated as

$$\hat{T}_{\text{prop}} = \frac{R_a R_b - D_a D_b}{R_a + D_a + R_b + D_b}, \quad (1)$$

where R_a, R_b are round-trip intervals and D_a, D_b are reply delays formed from DW3000 hardware timestamps. Antenna-delay calibration and the instantaneous carrier-integrator correction remain enabled. These are physical timestamp corrections, not temporal filtering.

2.2 Frame scheduler

The tag visits anchors A2, A3, A4, and A5 sequentially. For four anchors,

$$T_{\text{frame}} = 4T_{\text{slot}} + T_{\text{gap}}, \quad f_{\text{ideal}} = T_{\text{frame}}^{-1}. \quad (2)$$

Each complete position requires all four ranges associated with the same sequence round. The 64–21 ms profiles retain 2 ms response and final delays. The 17 and 13 ms profiles use 1 ms for each delayed transmission. Shortening these delays changes only the scheduled turnaround time, not the messages or the DS-TWR estimator.

Table 2: Timing profiles tested. All values except the ideal rate are in milliseconds.

Frame	Slot	Gap	RX timeout	RESP delay	FINAL delay	Ideal Hz
64	15	4	8	2	2	15.63
33	8	1	5	2	2	30.30
29	7	1	5	2	2	34.48
25	6	1	5	2	2	40.00
21	5	1	5	2	2	47.62
17	4	1	4	1	1	58.82
13	3	1	3	1	1	76.92

3 Experimental method

3.1 Geometry and reference data

The anchors form a nominal 3×3 m square and the tag is at (1.5, 1.5) m. Physical placement uncertainty is approximately ± 2 mm. The true center-to-corner range is $\sqrt{1.5^2 + 1.5^2} = 2.12132$ m. All equipment remained static and in line of sight throughout the sweep.

The 64 ms reference is the three-block DS-TWR data set from the preceding protocol comparison. It contains 360 s, 22,256 ranges, and 5,455 complete positions. Reusing it avoids redefining the baseline after seeing accelerated results.

3.2 Collection sequence

Every mode change was applied to all five modules in parallel and followed by a reboot and a warm-up interval. The eligible accelerated sequence was:

1. 60 s screening at 33, 29, 25, 21, 17, and 13 ms;
2. 120 s at 13 ms, then 120 s at 17 ms;
3. a second 120 s block at 13 ms, then a second at 17 ms.

The interleaved validation order limits the chance that temperature or slow environmental drift favors one final candidate.

3.3 Instrumentation controls

The first exploratory 33 ms run revealed a deterministic instrumentation artifact: extended DW3000 diagnostic-register reads and verbose log output executed periodically in the anchor’s real-time path. They caused the next poll to be missed in an almost exact five-sequence pattern. The firmware was changed once so that continuous positioning retains one compact result record per successful exchange while extended diagnostics remain available in the dedicated distance-test and survey modes. No protocol frame or ranging equation changed. Repeating the identical 33 ms profile improved frame completeness from 19.41% to 98.62%.

At 21 ms, the original collector reached a separate observation limit because it read only the latest distance per anchor from an HTTP snapshot. The radio still reported 99.982% successful exchanges. Collection was changed to an ordered log cursor, then the same profile was repeated. The valid repeat captured 99.89% complete frames.

Both exploratory runs are retained verbatim and marked `comparison_eligible=false`. Their exclusion was based on known instrumentation/collection behavior, not on range or position error.

3.4 Metrics

Ranges are grouped into rounds using the anchor index and the 16-bit sequence number. A frame is complete only if all four anchors are present. Positions are reconstructed by the same unfiltered linear trilateration used for the reference report. The reported error is Euclidean distance from (1.5, 1.5) m.

Radio success, timeouts, maximum exchange duration, and slot overruns come from one-second tag summaries and failure records. The historical baseline capture predates this timing-log collector, so its radio-success entry is shown as unavailable rather than inferred.

4 Results

4.1 Throughput and completeness

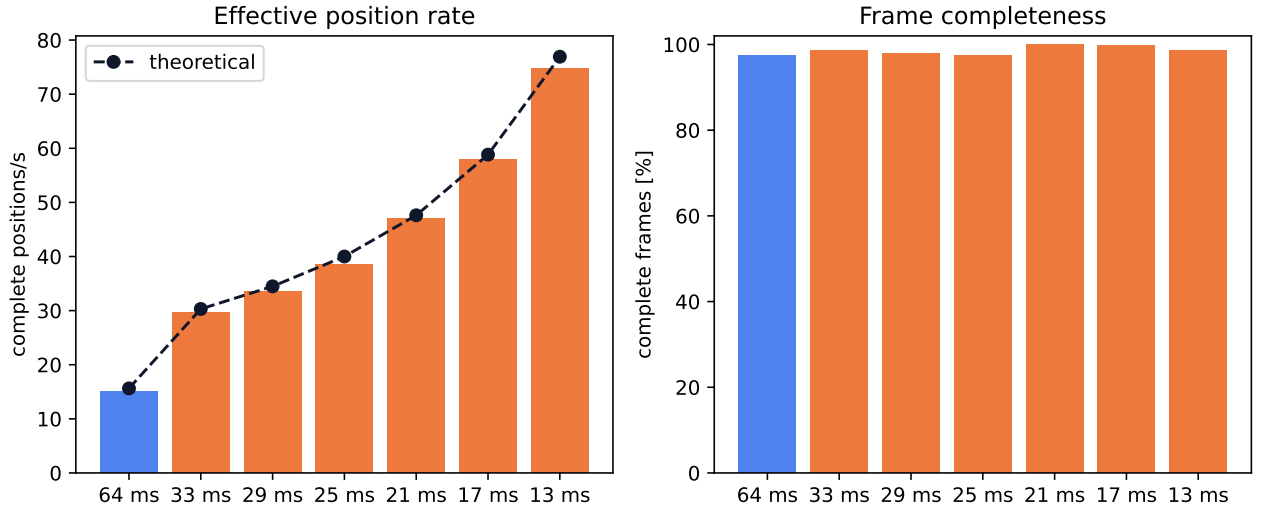


Figure 1: Effective complete-position rate versus the theoretical scheduler rate, and frame completeness. Final 17 and 13 ms values use validation blocks; intermediate profiles are screening results.

The 17 ms profile reaches 98.6% of its theoretical rate and produces 3.83 times the reference update rate. The 13 ms profile reaches 97.1% of theoretical and produces 4.93 times the reference update rate. Even at the tested limit, its 98.71% completeness remains above the 97.34% historical reference value.

4.2 Static accuracy

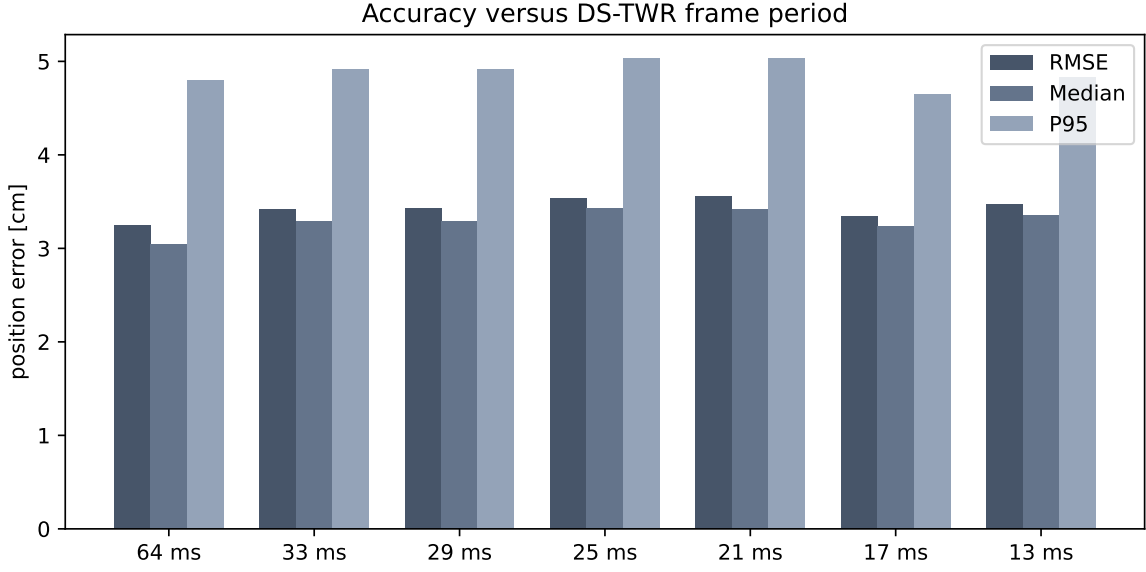


Figure 2: Unfiltered position RMSE, median, and P95 across the timing sweep.

There is no monotonic loss of accuracy with update rate. Relative to the reference, the validated 17 ms profile changes RMSE by only +0.99 mm and improves P95 by 1.45 mm. The 13 ms profile changes RMSE by +2.17 mm and P95 by +0.33 mm. These differences are comparable to placement uncertainty and between-block variation and should not be interpreted as a causal accuracy change.

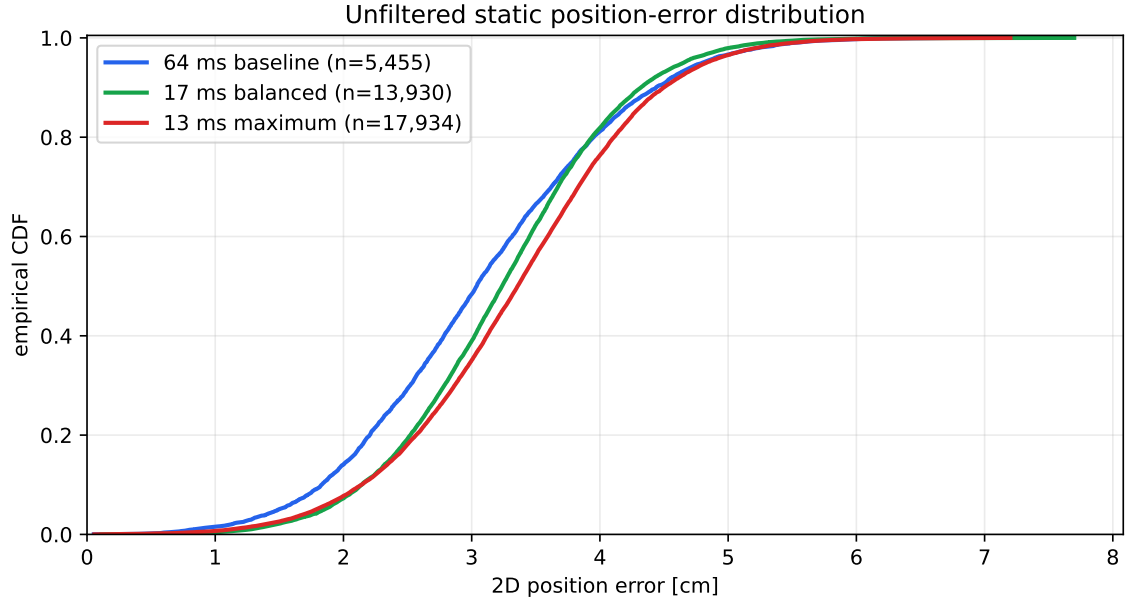


Figure 3: Empirical CDF of individual unfiltered position errors for the reference and the two validated accelerated profiles.

The distributions substantially overlap. The maximum errors were 6.79 cm at 64 ms, 7.70 cm at 17 ms, and 7.20 cm at 13 ms; no large outlier comparable to the previously excluded pathological FlexTDOA event appears.

4.3 Reliability and timing margin

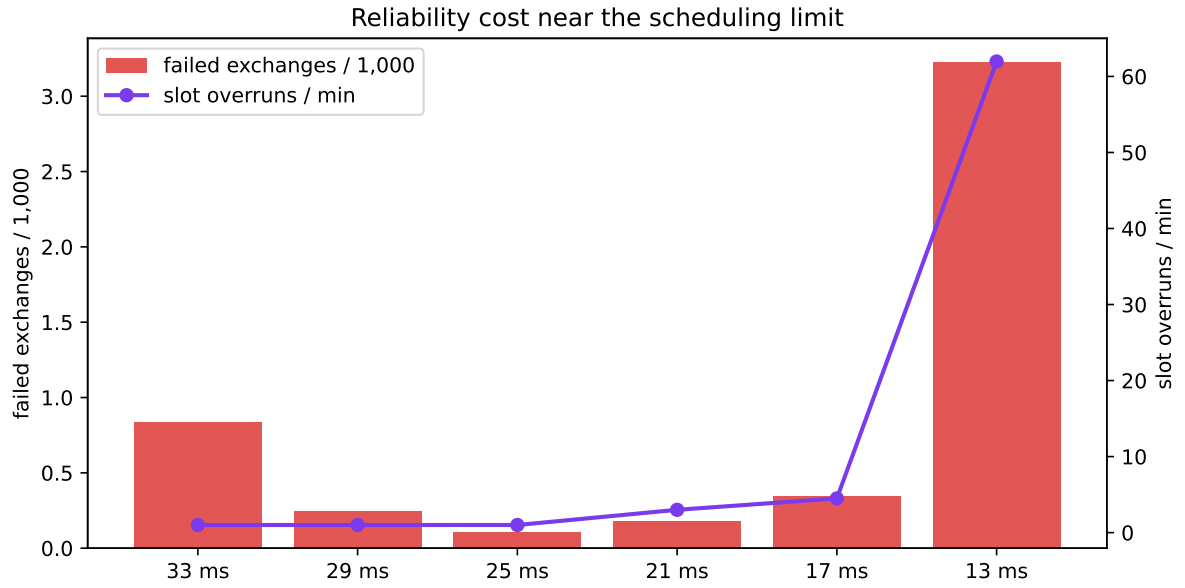


Figure 4: Failed exchanges and slot overruns. The final two profiles are derived from 240 s each; intermediate profiles are 60 s screening blocks.

Table 3: Reliability of the final candidates, pooled over two 120 s blocks.

Profile	Attempts	Failures	Success	Failures/1,000	Overruns/min
17 ms	55,640	19	99.966%	0.341	4.50
13 ms	72,604	234	99.678%	3.223	61.96

The faster profile gains 16.66 complete positions/s, but failed exchanges are about 9.4 times more frequent and slot overruns about 13.8 times more frequent. This is the clearest evidence that 13 ms is a boundary profile rather than a comfortable operating point.

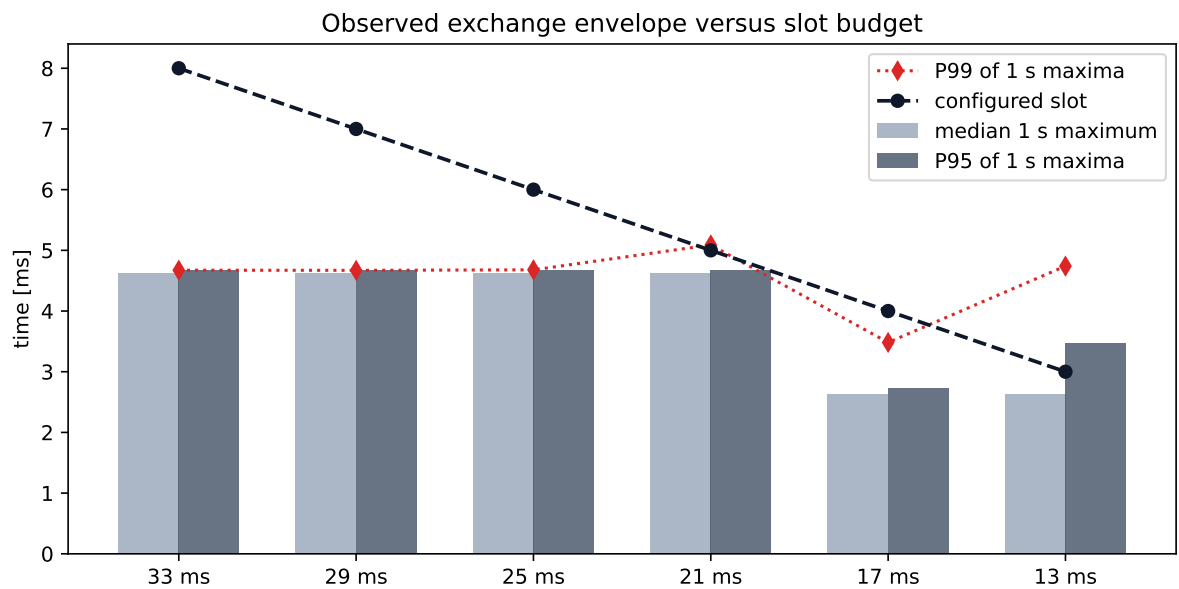


Figure 5: Distribution of the maximum exchange duration in failure-free one-second windows, compared with the configured slot budget.

At 17 ms, the median and P95 one-second maxima are 2.63 and 2.73 ms inside a 4 ms slot; even P99 is 3.48 ms. At 13 ms, the median is similar at 2.62 ms, but P95 rises to 3.46 ms and P99 to 4.74 ms, both beyond the 3 ms slot. Occasional scheduling/logging jitter therefore creates frequent overruns at the limit. A shorter nominal frame with the current integer-millisecond configuration would overrun by construction and was not treated as a valid profile.

4.4 Range bias and repeatability

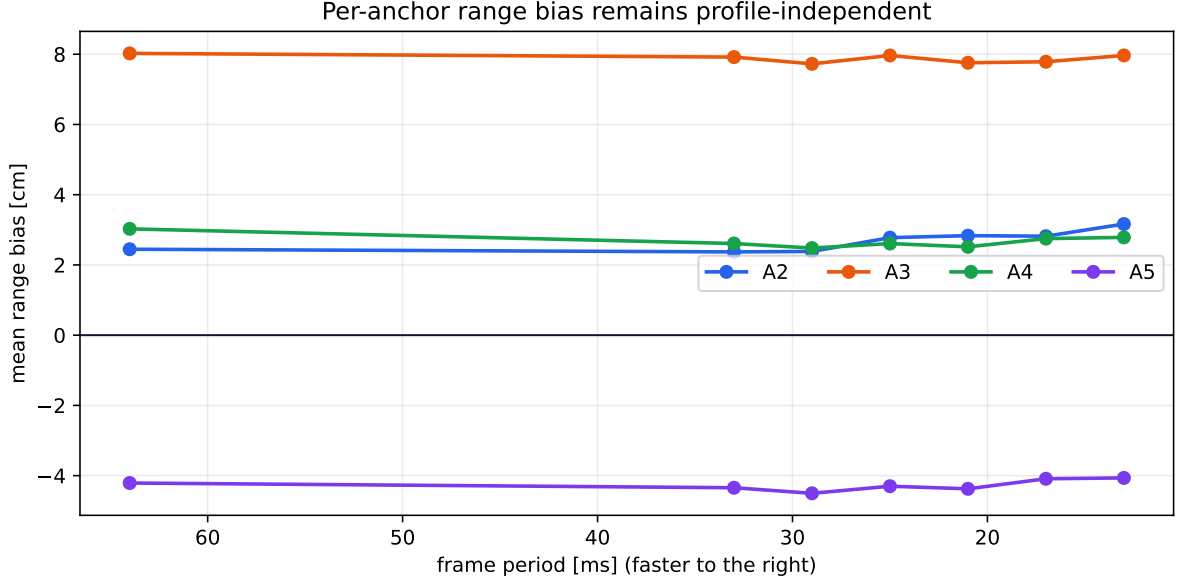


Figure 6: Per-anchor mean range bias over the sweep. The persistent module-dependent calibration offsets dominate any timing-profile effect.

Anchor biases remain approximately +2.4 to +3.2 cm for A2, +7.7 to +8.0 cm for A3, +2.5 to +3.0 cm for A4, and -4.5 to -4.1 cm for A5. Their stability explains why accelerated position accuracy stays close to the reference: response-delay reduction does not introduce a new systematic range shift in this data.

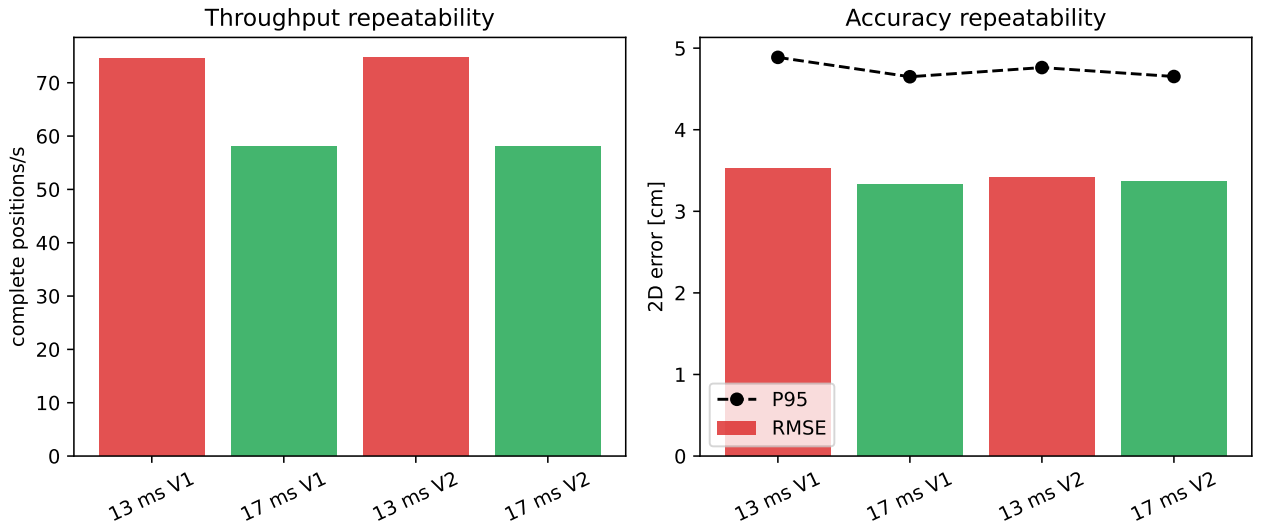


Figure 7: The two interleaved 120 s validation blocks for each final profile.

The validation blocks reproduce both rate and accuracy. The 13 ms position rates are 74.60 and

74.76 Hz; the 17 ms rates are 58.01 and 58.03 Hz. RMSE varies by 1.10 mm between the two 13 ms blocks and by 0.27 mm between the two 17 ms blocks.

5 Interpretation and recommendation

5.1 What limits the current implementation

The initial 64 ms configuration was dominated by deliberate idle time, not UWB airtime. Removing nonessential diagnostics from the real-time path and reducing scheduled turnaround delays exposes the actual implementation envelope:

- with 2+2 ms delayed transmissions, 21 ms is a strong operating point at 47.10 positions/s;
- with 1+1 ms delayed transmissions, 17 ms is robust and 13 ms is the current scheduling boundary;
- the 13 ms boundary is set by the 3 ms host slot, integer-millisecond runtime configuration, and occasional ESP32 task/log jitter—not by a demonstrated degradation of the DS-TWR distance estimator.

Moving below 13 ms cleanly would require sub-millisecond runtime timing, tighter real-time scheduling, or a design that pipelines work differently. That would be a separate firmware optimization study. Simply configuring a 2 ms slot today would make most healthy exchanges overrun their nominal slot and would not constitute a valid 9 ms frame.

5.2 Recommended profiles

Default: 17 ms balanced profile. Use slot 4 ms, round gap 1 ms, RX timeout 4 ms, response delay 1 ms, final delay 1 ms, and auto-RX delay 500 UUS. It gives 58.02 complete positions/s, 99.85% completeness, and accuracy statistically indistinguishable from the 64 ms reference, while retaining useful scheduling reserve.

Maximum-throughput option: 13 ms. Use slot 3 ms, gap 1 ms, timeout 3 ms, and 1+1 ms delayed transmissions when 74.68 positions/s is worth the reduced timing margin. The test remained accurate and 98.71% complete, but this setting should be revalidated under network load, motion, and longer operation before deployment.

Conservative option: 21 ms. If retaining the original 2+2 ms UWB reply delays is desirable, the 21 ms profile supplies 47.10 positions/s and 99.89% frame completeness.

6 Limits of inference and next tests

This is a rigorous speed sweep at one favorable static point, not a complete localization qualification. The conclusions apply to the tested modules, firmware, geometry, channel, and line-of-sight environment. They do not yet establish performance:

- near edges or outside the anchor square;
- while the tag is moving or rotating;
- under human blockage, multipath, or NLOS;
- with different anchor/tag antenna orientations;
- under heavy simultaneous Wi-Fi/dashboard load;
- over hours of thermal drift or with multiple tags.

The next study should use a spatial grid with independently measured ground truth, repeat 64/21/17/13 ms in randomized order at every point, and add controlled dynamic trajectories. The 17 ms profile is the appropriate starting point for those tests.

A Complete profile table

Table 4: Aggregate comparison. Intermediate profiles are screening blocks; 17 and 13 ms use validation data.

Frame	Positions/s	Complete	RMSE	Median	P95	Success	Fail/1,000
64	15.15	97.34%	3.25 cm	3.04 cm	4.80 cm	n/a	n/a
33	29.67	98.62%	3.41 cm	3.29 cm	4.92 cm	99.916%	0.835
29	33.46	97.81%	3.43 cm	3.28 cm	4.92 cm	99.976%	0.242
25	38.61	97.44%	3.53 cm	3.43 cm	5.03 cm	99.989%	0.106
21	47.10	99.89%	3.56 cm	3.42 cm	5.03 cm	99.982%	0.177
17	58.02	99.85%	3.35 cm	3.24 cm	4.65 cm	99.966%	0.341
13	74.68	98.71%	3.47 cm	3.36 cm	4.83 cm	99.678%	3.223

B Files and reproduction

File/directory	Contents
data/*.jsonl.xz	lossless archives of raw range/status events and timing logs
data/*.metadata.json	timing configuration, module state, and collection metadata
raw_data_manifest.csv	byte counts and SHA-256 checksums for raw and archived files
block_metrics.csv	metrics for every baseline, screening, validation, and exploratory block
profile_metrics.csv	phase-aware aggregate metrics
positions.csv	every reconstructed complete position
analysis_summary.json	machine-readable analysis output
figures/	all plots in vector PDF and PNG

Reproduction

```
cd /home/pi/Documents/UWB
xz --decompress --keep reports/ds_twr_speed_limit_20260726/data/*.jsonl.xz
python3 tools/uwb_ds_speed_analyze.py
cd reports/ds_twr_speed_limit_20260726
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The analysis never modifies raw event files. The original 64 ms blocks live under `reports/uwb_protocol_comparison_20260726/data`; the path is recorded in `analysis_summary.json`.

References

- [1] Qorvo, *DW3xxx Device Driver API Guide*, double-sided two-way-ranging examples, 2025. <https://forum.qorvo.com/uploads/short-url/xD3TlXKvkujjdUaJWxv2b4E7GQN.pdf>.
- [2] Decawave, *APS013: DW1000 and Two-Way Ranging*, rev. 2.3, Poll-Response-Final and the asymmetric DS-TWR formula. <https://forum.qorvo.com/uploads/short-url/x34DrF7EW5fQP9wY3aNESqPKz8z.pdf>.
- [3] Qorvo, *DW3110 Ultra-Wideband Transceiver*, product documentation. <https://www.qorvo.com/products/p/DW3110>.
- [4] Local repository *UWB Module ESP-IDF*, branch `experiment/ds-twr-native-performance`, reference commit 27a28b03f1dd627cdcbbf327fb94127bb3fe371b, working state from July 26, 2026.